Lab experiment 8

Develop Inventory Management Module using ADO.NET & CRUD Operations

Select – Recent -ASP.NET Framework
Give Project Name – InventoryManagementModule

Create a new ASP.NET Web Application

Empty

An empty project template for creating ASP.NET applications. This template does not have any content in it.

Web Forms

A project template for creating ASP.NET Web Forms applications. ASP.NET Web Forms lets you build dynamic websites using a familiar drag-and-drop, event-driven model. A design surface and hundreds of controls and components let you rapidly build sophisticated, powerful UI-driven sites with data access.

MVC

A project template for creating ASP.NET MVC applications. ASP.NET MVC allows you to build applications using the Model-View-Controller architecture. ASP.NET MVC includes many features that enable fast, test-driven development for creating applications that use the latest standards.

Web API

A project template for creating RESTful HTTP services that can reach a broad range of clients including browsers and mobile devices.

Single Page Application

A project template for creating rich client side JavaScript driven HTML5 applications using ASP.NET Web API. Single Page Applications provide a rich user experience which includes client-side interactions using HTML5, CSS3, and JavaScript.

Authentication

None

Add folders & core references

- [ ] Web Forms
- [x] MVC
- [ ] Web API

Advanced

- [x] Configure for HTTPS
- [ ] Container support

(Requires Docker Desktop)

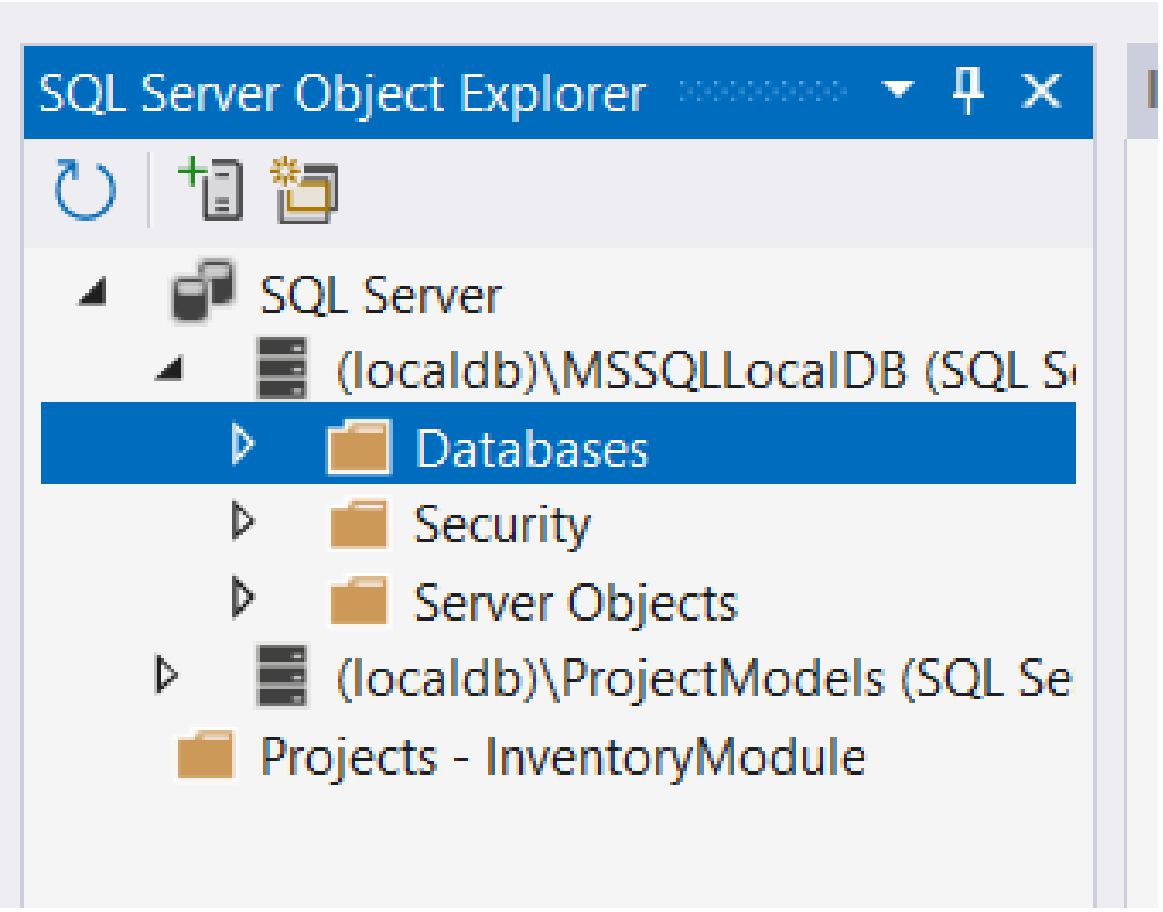
- [ ] Also create a project for unit tests

InventoryModule.Tests

Back Create

From - View Tab

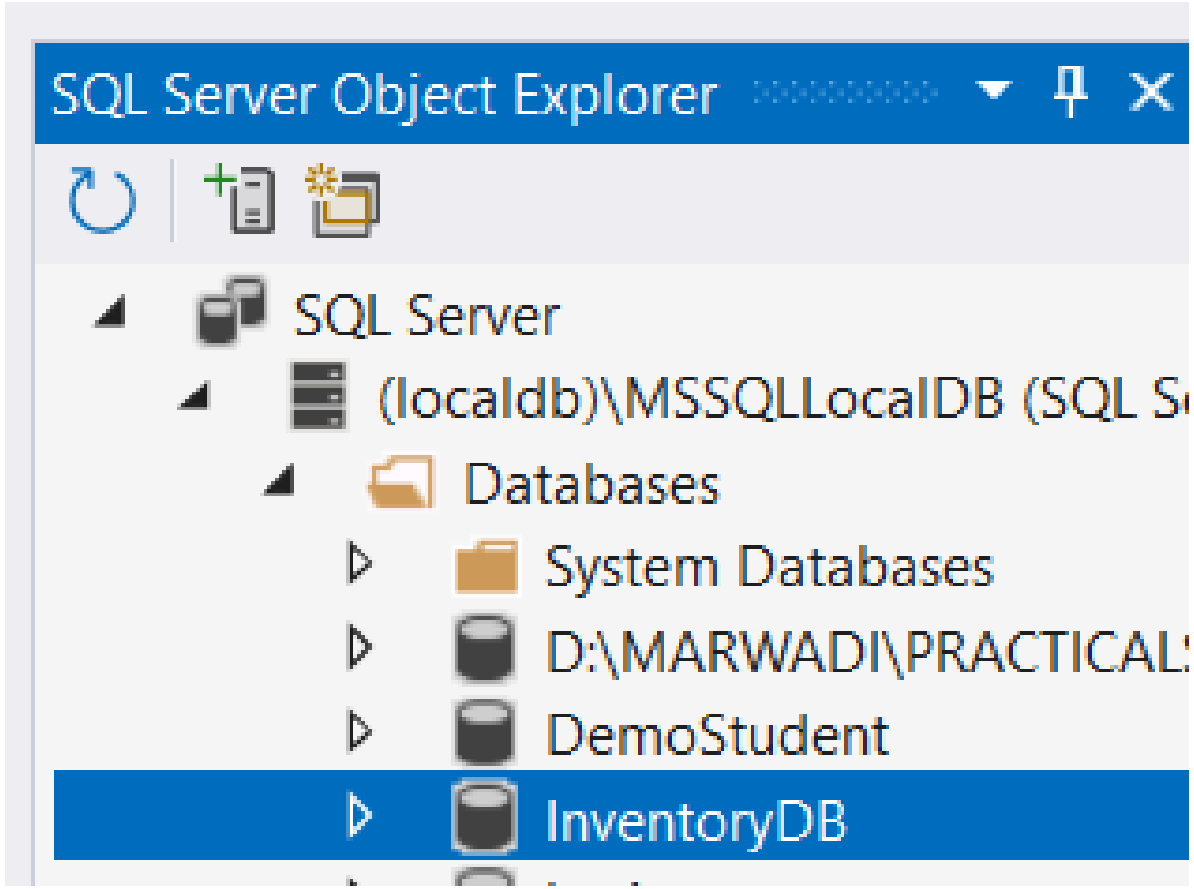
Click - SQL Server Object Explorer



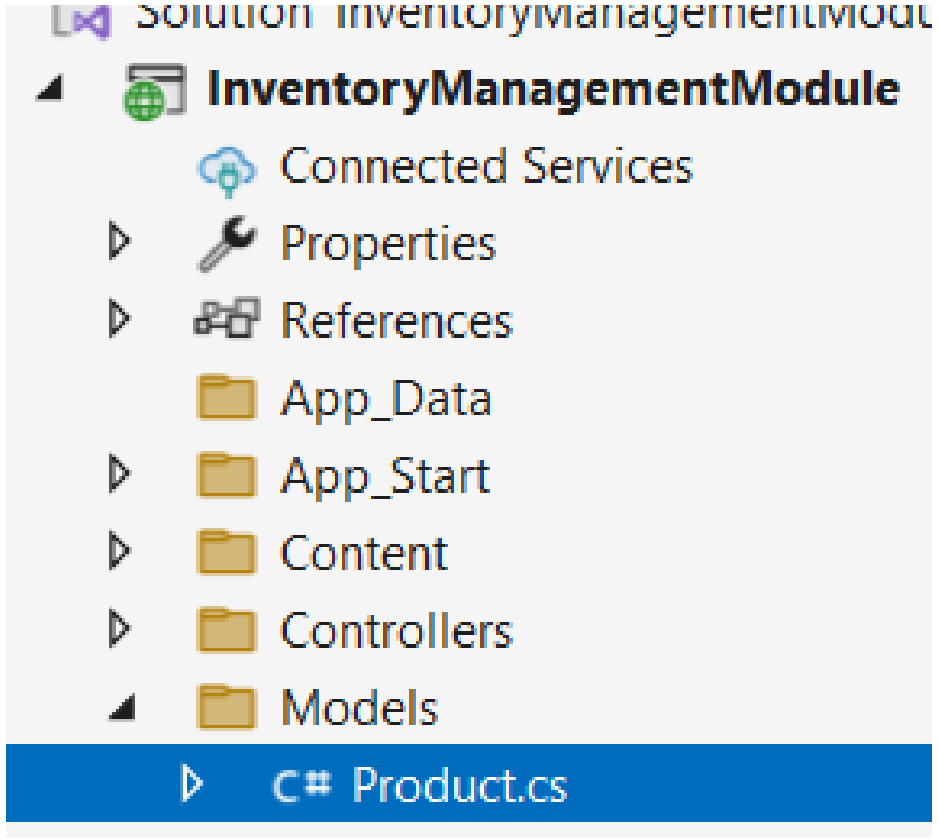
Select Database

Right Click – Add New DataBase

Give the Name - InventoryDB



Select InventroyDB
In dropdown – Select Tables
Right Click – Add New Table
Products



Select Models Folder –
Right Click – Add – Class
Give the name - Product

```
namespace InventoryManagementModule.Models
{
    7 references
    public class Product
    {
        3 references
        public int ProductId { get; set; }
        4 references
        public string ProductName { get; set; }
        4 references
        public int Quantity { get; set; }
        4 references
        public decimal Price { get; set; }
    }
}
```

Controller – ProductController

```
public class ProductController : Controller
{
    string cs = "Data Source=(localdb)\\MSSQLLocalDB;Initial Catalog=InventoryDB;Integrated Security=True";
    // GET: Product
    0 references
    public ActionResult Index()
    {
        List<Product> products = new List<Product>();
        using (SqlConnection con = new SqlConnection(cs))
        {
            string query = "SELECT * FROM Products";
            SqlCommand cmd = new SqlCommand(query, con);
            con.Open();
            SqlDataReader dr = cmd.ExecuteReader();
            while (dr.Read())
            {
                products.Add(new Product
                {
                    ProductId = Convert.ToInt32(dr["ProductId"]),
                    ProductName = dr["ProductName"].ToString(),
                    Quantity = Convert.ToInt32(dr["Quantity"]),
                    Price = Convert.ToDecimal(dr["Price"])
                });
            } }return View(products);
    }
```

```
public ActionResult Create()
{       return View();          }
//CREATE : Save Product
[HttpPost]
0 references
public ActionResult Create(Product p)
{
    using (SqlConnection con = new SqlConnection(cs))
    {
        string query = @"INSERT INTO Products
                (ProductId,ProductName,Quantity,Price)
        VALUES(@ProductId,@ProductName,@Quantity,@Price)";

        SqlCommand cmd = new SqlCommand(query,con);
        cmd.Parameters.AddWithValue("@ProductId",p.ProductID);
        cmd.Parameters.AddWithValue("@ProductName", p.ProductName);
        cmd.Parameters.AddWithValue("@Quantity", p.Quantity);
        cmd.Parameters.AddWithValue("@Price", p.Price);
        con.Open();
        cmd.ExecuteNonQuery();
    }
    return RedirectToAction("Index");
}
```

```
//UPDATE : Display existing Product
0 references
public ActionResult Edit(int id)
{
    Product p = new Product();
    using (SqlConnection con = new SqlConnection(cs))
    {
        string query = "SELECT * FROM Products WHERE " +
                       "ProductId=@ProductId";
        SqlCommand cmd = new SqlCommand(query, con);
        cmd.Parameters.AddWithValue("@ProductId", id);
        con.Open();
        SqlDataReader dr = cmd.ExecuteReader();
        if (dr.Read())
        {
            p.ProductID = Convert.ToInt32(dr["ProductId"]);
            p.ProductName = dr["ProductName"].ToString();
            p.Quantity = Convert.ToInt32(dr["Quantity"]);
            p.Price = Convert.ToDecimal(dr["Price"]);
        }
        else { return HttpNotFound(); }
    }return View(p);
}
```

```
//UPDATE : Save Changes
0 references
public ActionResult Edit(Product p)
{
    using (SqlConnection con = new SqlConnection(cs))
    {
        string query = @"UPDATE Products SET
                        ProductName=@ProductName,
                        Quantity=@Quantity,
                        Price=@Price
                        WHERE ProductId=@ProductId";
        SqlCommand cmd=new SqlCommand(query, con);
        cmd.Parameters.AddWithValue("@ProductId", p.ProductID);
        cmd.Parameters.AddWithValue("@ProductName", p.ProductName);
        cmd.Parameters.AddWithValue("@Quantity", p.Quantity);
        cmd.Parameters.AddWithValue("@Price", p.Price);
        con.Open();
        cmd.ExecuteNonQuery();
    }
    return RedirectToAction("Index");
}
}
```

```
//DELETE : Remove product
[HttpPost]
0 references
public ActionResult Delete(int id)
{
    using (SqlConnection con = new SqlConnection(cs))
    {
        string query = "DELETE FROM Products WHERE ProductId=@ProductId";
        SqlCommand cmd = new SqlCommand(query, con);
        cmd.Parameters.AddWithValue("@ProductId", id);
        con.Open();
        cmd.ExecuteNonQuery();
    }
    return RedirectToAction("Index");
}
```

Under Views Folder – Select Product Folder – Right click – Add

Views

Give the name - Create

```
@model InventoryManagementModule.Models.Product
@{
    Layout = null;
}
<h2>Add New Product</h2>

@using (Html.BeginForm())
{
    <p>Product Name:</p>
    @Html.TextBoxFor(m => m.ProductName)
    <br />    <br />

     <p>Quantity:</p>
    @Html.TextBoxFor(m => m.Quantity)
    <br />    <br />

    <p>Price:</p>
    @Html.TextBoxFor(m => m.Price)
    <br />     <br />

    <input type="submit" value="Save Product" />
}
<br />
@Html.ActionLink("Back to Inventory", "Index")
```

Name of your Project/Namespace

Create.cshtml

Under Views Folder – Select Product Folder – Right click – Add

Views

Give the name - Edit

```
@model InventoryManagementModule.Models.Product
@{
    Layout = null;
}
<h2>Edit Product</h2>
@using (Html.BeginForm())
{
    @Html.HiddenFor(m => m.ProductId)
    <p>Product Name:</p>
    @Html.TextBoxFor(m => m.ProductName)
    <br />    <br />

    <p>Quantity:</p>
    @Html.TextBoxFor(m => m.Quantity)
    <br />    <br />

    <p>Price:</p>
    @Html.TextBoxFor(m => m.Price)
    <br />    <br />

    <input type="submit" value="Update Product" />
}
<br />
@Html.ActionLink("Back to Inventory", "Index")
```

Edit.cshtml

Under Views Folder – Select Product Folder – Right click – Add

Views

Give the name - Index

```
@model List<InventoryManagementModule.Models.Product>
@{      Layout = null;    }
<h2>Inventory Management</h2>
<p>     @Html.ActionLink("Add New Product", "Create")</p>
<table border="1">
    <tr>
        <th>Product ID</th>
        <th>Product Name</th>
        <th>Quantity</th>
        <th>Price</th>
        <th>Actions</th>
    </tr>
    @foreach (var item in Model)
    {
        <tr>
            <td>@item.ProductId</td>
            <td>@item.ProductName</td>
            <td>@item.Quantity</td>
            <td>@item.Price</td>
            <td>
                @Html.ActionLink("Edit", "Edit",new { id = item.ProductId })
                @using (Html.BeginForm("Delete", "Product",new { id = item.ProductId },
                 FormMethod.Post))
                {       <input type="submit" value="Delete" />      }
            </td>  </tr>     }
</table>
```

Index.cshtml

```
routes.IgnoreRoute("{resource}.axd/{*pathInfo}");

routes.MapRoute(
    name: "Default",
    url: "{controller}/{action}/{id}",
    defaults: new { controller = "Product",
        action = "Index", id = UrlParameter.Optional }
);
```

In RouteConfig.cs file

Replace the name from Home to Product